Cryptocurrency Trend Classification Across Multiple Assets Using Machine Learning

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***Abstract*—//**
This paper presents a machine learning study for cryptocurrency price movement classification using data collected from Binance API at 4-hour intervals from 2020 to 2026. The dataset includes eight major cryptocurrencies: BTC, ETH, BNB, SOL, ADA, XRP, DOGE, and AVAX. The target is defined as a three-class classification problem: Bigup, Stable, and Bigdown, based on the next 4-hour return calculated separately for each coin. A feature engineering pipeline was applied to capture market behavior, including price movement, volatility, momentum, volume activity, taker buy/sell ratios, and price position within the candle. The main experiment follows a cross-asset pattern learning approach, where models are trained on several coins and tested on other coins to examine whether movement patterns can transfer across cryptocurrencies. Five machine learning models were evaluated: Random Forest, XGBoost, SVM, Logistic Regression, and KNN. The results showed that Random Forest was the most reliable model, while ADA and AVAX showed stronger predictive behavior. Therefore, a secondary chronological forecasting case study was conducted on ADA and AVAX, where Random Forest achieved 61 test accuracy and 0.61 macro F1-score, outperforming the LSTM model.

***Index Terms*—cryptocurrency, machine learning, price trend prediction, feature engineering, Random Forest, XGBoost, SVM, Logistic Regression, KNN, LSTM, 4-hour candle, multi-coin, Binance**

I. INTRODUCTION

The global cryptocurrency market has grown into a major financial asset class that operates continuously, 24 hours a day and 7 days a week. Unlike traditional equity markets, cryptocurrencies experience sharp price movements due to factors such as supply and demand, trading activity, social media sentiment, regulatory news, technical momentum, and speculative behavior. This high volatility makes cryptocurrencies attractive for traders, but also difficult to analyze and predict. Predicting cryptocurrency price movement is one of the challenging tasks in financial machine learning. Exact price prediction is difficult because cryptocurrency prices are noisy and can change significantly within short periods. Therefore, predicting the direction of movement can be more practical than predicting the exact price value. In this study, the prediction task is formulated as a classification problem, where each price movement is classified as Bigup, Stable, or Bigdown. Previous studies have shown that machine learning models can learn useful patterns from cryptocurrency data. Some studies focused on exact price forecasting using models such as Random Forest, Gradient Boosting, Neural Networks, and LSTM. Other studies focused on trend classification, where the goal is to predict whether the market will move upward, downward, or sideways. For example, Silva et al. proposed a Bitcoin trend prediction framework using feature engineering based on temporal variables and technical indicators, and their results showed that tree-based models such as Random Forest can achieve strong performance in specific high-frequency Bitcoin trend classification settings. There is also an important discussion in the literature about whether complex deep learning models are always better than traditional machine learning models. Some studies suggest that when the data is structured as engineered tabular features, tree-based models such as Random Forest and XGBoost can perform competitively or even better than deep learning models. This is important for cryptocurrency prediction because feature engineering may be as important as model complexity. This study focuses on cross-asset cryptocurrency price movement classification. Instead of studying only one cryptocurrency, the project investigates whether movement patterns learned from some coins can transfer to other coins. The dataset contains 4-hour Binance candles from 2020 to 2026 for eight cryptocurrencies: BTC, ETH, BNB, SOL, ADA, XRP, DOGE, and AVAX. A feature engineering pipeline is applied to describe price behavior, trend, volatility, momentum, volume activity, taker buy/sell pressure, and price position. The main experiment evaluates cross-asset pattern learning using five supervised machine learning models: Random Forest, XGBoost, Support Vector Machine, Logistic Regression, and K-Nearest Neighbors. In addition, a secondary forecasting case study is conducted on ADA and AVAX using chronological splitting, where Random Forest and LSTM are compared. This allows the study to examine both shared market behavior across cryptocurrencies and time-based forecasting performance on selected coins. The rest of this paper is organized as follows. Section 2 reviews related work. Section 3 presents the methodology, including data collection, preprocessing, feature engineering, target definition, experiments, models, and evaluation metrics. Section 4 reports the experimental results. Section 5 discusses the findings and limitations. Section 6 concludes the paper and outlines future work.

II. Related Work

A. Machine Learning for Cryptocurrency Price Prediction

Alnami et al. (2025) presented an integrated framework for cryptocurrency price forecasting and anomaly detection using Random Forest, Gradient Boosting, and feedforward neural networks across Bitcoin, Ethereum, Binance Coin, and Litecoin. Their study demonstrated that Random Forest consistently achieved the highest predictive accuracy, with R-squared values above 0.999, and showed that a Z-Score-based anomaly detection framework could reliably classify market events as normal or abnormal. The study highlights the robustness of ensemble learning methods when applied to structured price data with engineered features such as trading volume and market capitalisation. Silva et al. (2025) conducted a benchmark comparison of over 20 machine learning models for Bitcoin trend prediction at 5-minute, 15-minute, and 1-hour granularities. Their key finding was that a carefully designed feature engineering approach combining temporal variables (hour, day of week, day of year) with recalibrated technical indicators such as SMA, EMA, RSI, MACD, and Bollinger Bands significantly outperformed two established benchmark frameworks by up to 73 in relative accuracy. Their top result was a Random Forest model achieving 91.9 accuracy at 5-minute intervals. Crucially, they found that deep learning models such as TFT and CNN-LSTM consistently underperformed tree-based ensembles when working with tabular engineered features, reinforcing the idea that feature quality matters more than model complexity. Cakici et al. (2024) studied the cross-section of cryptocurrency returns using machine learning and found that simpler methods, particularly ordinary least squares, outperformed complex tree-based and neural network methods in predicting future returns across many coins. Their study across over 1,500 cryptocurrencies showed that the benefits of model complexity seen in stock markets do not carry over directly to crypto markets, and that lagged momentum, size, and value-related features are the strongest predictors of cross-sectional returns. This finding strongly supports the choice of interpretable, well-tuned tree-based models over deep learning when working with tabular cryptocurrency features. Zhang et al. (2025) used machine learning to classify price time series and distinguish between cryptocurrencies and stocks. Using features such as mean, variance, skewness, kurtosis, and autocorrelation, they showed that the price patterns of cryptocurrencies are statistically distinct from those of stocks, reflecting different investor behaviour. The study found that logistic regression, random forest, and SVM achieved very high classification accuracy in separating the two asset classes. This is directly relevant to our work because it confirms that cryptocurrency price series have learnable and distinctive statistical properties that machine learning models can exploit

B. Feature Engineering for Trend Classification

Feature engineering is consistently identified as one of the most important steps in cryptocurrency forecasting. Silva et al. (2025) demonstrate that temporal variables such as the hour of day carry surprisingly strong predictive power, particularly at fine time granularities. At 15-minute intervals, technical indicators such as the MACD signal line and Bollinger Bands emerge as the most predictive features, with balanced importance across all engineered features. Multiple studies confirm that moving averages, volume ratios, and momentum indicators provide meaningful signals. The z-score, which measures how far the current price is from its recent rolling average in units of standard deviation, is a particularly useful feature for identifying unusual price levels. Volume spike ratios, which compare today's volume to the rolling average, are effective for detecting candles that may precede large price moves. The taker buy volume which we calculate from taker buy ratio and taker sell ratio, a feature available from exchange data such as Binance, captures the proportion of trades that were aggressively executed by buyers. A high taker buy ratio signals bullish pressure while a low ratio signals selling pressure. This feature is not available in most academic datasets and represents an advantage of using directly sourced exchange data.

C. Deep Learning vs. Ensemble Methods

There is an ongoing debate about whether deep learning models such as LSTM outperform classical ensemble methods for price prediction. LSTM networks are designed to capture temporal dependencies in sequential data, which makes them theoretically well-suited to time series prediction. However, the empirical evidence from multiple studies suggests that when the input is a well-engineered set of tabular features, tree-based methods are more effective. Silva et al. found that TFT and CNN-LSTM underperformed tree-based ensemble methods in their experimental setup. Alnami et al. (2025) similarly found that deep learning models achieved slightly higher error metrics than Random Forest and Gradient Boosting on most cryptocurrency datasets. Our study tests this finding directly by including an LSTM model alongside tree-based methods and comparing performance across the same evaluation conditions.

III. Data Description

A. Source and Granularity

All price data used in this study were sourced from Binance, the world's largest cryptocurrency exchange by trading volume. Data were collected at a 4-hour candle granularity, meaning each row in the dataset represents a 4-hour trading period. The choice of 4-hour candles represents a balance between detail and noise. Shorter candles such as 1-minute or 5-minute carry a great deal of microstructure noise and random fluctuations that make signal extraction difficult. Longer candles such as daily data smooth out important intraday patterns. The 4-hour interval is widely used by professional cryptocurrency traders and aligns with major trading sessions across different global time zones. Eight coins were selected to represent a diverse range of cryptocurrency types: Bitcoin (BTC), the largest coin by market capitalisation; Ethereum (ETH), the leading smart contract platform; Binance Coin (BNB), an exchange utility token; Solana (SOL), a high-performance layer-1 blockchain; Cardano (ADA), an academically designed proof-of-stake coin; XRP, a payment-focused digital asset; Dogecoin (DOGE), a meme-originated coin with high retail trader interest; and Avalanche (AVAX), a smart contract platform with a unique consensus mechanism. This selection covers large-cap, mid-cap, utility, speculative, and platform coins, providing a diverse testing ground.

B. Raw Features

Each raw dataset contains 10 columns per candle. The date column records the timestamp of the candle opening. The open column is the price at which the candle began. The high and low columns record the maximum and minimum prices reached during the 4-hour period. The close column is the final price of the candle. The volume column records how many coins changed hands during the period. The quotevolume column gives the equivalent value in US dollars of all trades. An additional column, takerbuyvolume, records the volume that was executed by aggressive market buyers, which is used to compute the taker buy ratio , numtrades it is number of trades that happen in that day

C. Dataset Size

Each coin dataset originally contains approximately 13907 rows, corresponding to roughly 365 days of 4-hour candles for six years when accounting for the fact that cryptocurrency markets operate continuously. After feature engineering, the first 30 rows of each dataset are dropped because rolling window features with a 30-period window are undefined for earlier rows. After cleaning and dropping rows with any remaining missing values, When all six coins used in modelling (ETH, BNB, SOL, ADA, DOGE, AVAX) are combined, BTC and XRP are dropped before modelling for the reasons discussed in Section 5.

IV. Feature Engineering

A. Candle Body

The candle body is computed as the difference between the close price and the open price of each candle. A positive candle body means the price went up during the 4-hour period, which is a bullish signal. A negative candle body means the price fell, which is a bearish signal. The magnitude of the body indicates how strong the move was. In candlestick trading, a large green body suggests strong buying pressure, while a large red body suggests strong selling pressure.

B. High-Low Range

The high-low range is the difference between the highest and lowest price within the candle. This feature measures how volatile or how wide the price movement was during the 4-hour period. A large range suggests uncertainty or a strong directional move with significant price action. A small range indicates a calm, low-volatility candle. This feature is particularly useful because high-volatility candles often precede or follow major price moves.

C. Moving Averages

A moving average (MA) is the average closing price over the last N candles. The 7-period MA (MA7) captures the short-term trend over approximately 28 hours, while the 30-period MA (MA30) captures the medium-term trend over approximately 5 days. Moving averages are one of the oldest and most widely used technical indicators. When the price is above its moving average, the market is considered to be in an uptrend; when below, it is in a downtrend. The moving averages themselves are dropped as input features before modelling because they are in dollar terms, but they are used to compute ratios.

D. MA Ratios

The MA ratios express how far the current close price is from its moving average, as a proportion. For example, if close = 105 and MA7 = 100, then ma7ratio = 0.05, meaning the price is 5 above its 7-period average. A positive ratio means the price is above its average (overbought tendency) and a negative ratio means the price is below its average (oversold tendency). These ratios are scale-independent and can be compared meaningfully across different coins

E. Percentage Return

The moving1d feature is the percentage change from the previous candle's close price to the current candle's close price. This is computed using the pandas pctchange() function multiplied by 100. It tells the model how much the price moved in the most recent candle. For example, a value of 2.5 means the price rose 2.5 in the last 4 hours, while a value of -1.8 means the price fell 1.8

F. Volatility

Volatility features measure how much the daily percentage returns have been fluctuating recently. They are computed as the rolling standard deviation of moving1d over windows of 3, 7, and 14 candles. A high volatility value means the price has been jumping around a lot recently, which can signal an uncertain or trending market. A low volatility value means the market has been stable. Multiple windows are used to capture short-term (3-candle), medium-term (7-candle), and longer-term (14-candle) volatility regimes.Volatility features measure how much the daily percentage returns have been fluctuating recently. They are computed as the rolling standard deviation of moving1d over windows of 3, 7, and 14 candles. A high volatility value means the price has been jumping around a lot recently, which can signal an uncertain or trending market. A low volatility value means the market has been stable. Multiple windows are used to capture short-term (3-candle), medium-term (7-candle), and longer-term (14-candle) volatility regimes.

G. Volume Spike

The volume spike feature is the ratio of the current candle's volume to the 7-period rolling average volume. A value of 2.0 means twice the normal trading activity occurred in this candle. Volume spikes are important signals in technical analysis because unusually high volume often accompanies significant price moves. A volume spike combined with a large price return is typically a stronger signal than a large price return on normal volume..

H. Z-Score

The z-score measures how far the current closing price is from its 30-period rolling mean, expressed in units of the rolling standard deviation. A z-score near zero means the price is close to its recent average. A z-score of +2 means the price is two standard deviations above average, suggesting an unusually high price. A z-score of -2 suggests an unusually low price. This feature is useful because extreme z-scores often precede mean-reversion moves. The rolling mean and standard deviation columns used to compute the z-score are dropped after the z-score is created.

I. Next Return and Target Label

To create the target variable, the next candle's closing price is computed by shifting the close column by -1 row. The nextreturn is then computed as the percentage change from the current close to the next close. These two columns are dropped from the input features before modelling because they represent future information that would not be available at prediction time. The label column is derived from nextreturn using quantile thresholds explained in Section 5.

J. Taker Buy and Sell Ratios

The takerbuyratio is computed as the taker buy volume divided by total volume. A high ratio (close to 1.0) means most of the trades were executed by aggressive buyers, indicating bullish market pressure. A low ratio means most trades were executed by sellers, indicating bearish pressure. The takersellratio is simply 1 minus the taker buy ratio. Both features capture the balance of power between buyers and sellers during the candle, which is a direct measure of market microstructure

K. Price Position

Price position measures where the closing price falls within the candle's high-low range. It is computed as (close - low) / (high - low). A value close to 1.0 means the candle closed near its high, indicating buyer dominance and a bullish signal. A value close to 0.0 means the candle closed near its low, indicating seller dominance. This feature is related to the concept of the wick in candlestick analysis.

L. Momentum

The 7-candle momentum feature compares the current close price to the close price from 7 candles ago. It is computed as (close / close.shift(7)) - 1, giving the percentage return over the last 7 candles (28 hours). Positive momentum means the price has been trending upward over this window. Negative momentum means it has been falling. Momentum is one of the most studied predictive factors in both stock and cryptocurrency markets..

M. Negative Momentum and Down Days

Two additional bearish-focused features are created. The negativemomentum feature takes the value of momentum7 when it is negative and sets it to zero when momentum is positive. This isolates the downside momentum signal. The downdays7 feature counts how many of the last 7 candles had a negative percentage return. If six out of seven recent candles were red, the market is likely in a downtrend. These features help the model learn asymmetric patterns where downward moves behave differently from upward moves.

V. Target Label Creation and Class Distribution

A. Three-Class Classification

The prediction task in this study is a three-class classification problem. Rather than predicting the exact price or a binary up/down direction, we classify the next candle's return into one of three categories. This three-class approach provides more useful information than a binary label and allows the model to express uncertainty about the direction through the "Stable" class. The three labels are: Bigup, assigned to candles where the next return falls in the top 30 of all returns for that coin; Bigdown, assigned to candles where the next return falls in the bottom 30; and Stable, assigned to the middle 40. This labelling scheme uses quantile-based thresholds, meaning the thresholds adapt to each coin's own return distribution.

B. Why Quantile Thresholds

Using fixed thresholds such as "+3 is a Bigup" would be problematic because different coins have very different typical daily return magnitudes. Bitcoin, with its high absolute price, has relatively small typical percentage moves, while Dogecoin can regularly move 5-10 in a 4-hour period. By using the 30th and 70th percentiles, the threshold automatically adapts to each coin's own volatility profile, ensuring that each class contains approximately 30 of the data for Bigup and Bigdown, and 40 for Stable. This produces a near-balanced three-class distribution, which simplifies the modelling problem.

C. Class Balance

After applying the quantile thresholds, each coin's dataset has approximately 30 of samples labelled Bigup, 30 Bigdown, and 40 Stable. This distribution is intentionally designed to be near-balanced. A fully balanced three-class problem would have 33.3 in each class. The 30/40/30 split is slightly imbalanced toward Stable but is much more balanced than real-world financial data typically is. Despite the near-balance, models are trained with classweight=balanced to ensure the minority classes (Bigup and Bigdown) receive adequate attention during training.

D. Exclusion of BTC and XRP

BTC was excluded from the modelling stage because it often acts as a market leader and may dominate the movement patterns of other cryptocurrencies and also 2022 data was little have big effect which make it big noise . XRP is excluded due to its unique regulatory environment and behavioural characteristics that differ significantly from other major coins. Including these two coins in training would add patterns that do not generalise well to the remaining coins.

VI. Models and Evaluation Strategy

A. Input Features

Before training, the following columns are dropped from the input features: label (the target), date, type (coin identifier), downdays7, takerbuyvolume (the raw column, since the ratio is used instead), zscore, takerbuyratio, and negativemomentum. This leaves 13 input features: candlebody, highlowrange, ma7ratio, ma30ratio, moving1d, volatility7, volumespike, takersellratio, priceposition, momentum7, volatility14, volatility3,numtrades

B. Random Forest

Random Forest is an ensemble learning method that builds a large number of decision trees during training, each trained on a random bootstrap sample of the data with a random subset of features. The final prediction is determined by majority vote across all trees. The key advantage of Random Forest is that it reduces the overfitting problem common to individual decision trees by averaging across many uncorrelated trees. In this study, 700 trees are used with a maximum depth of 13, minimum samples per split of 5, minimum samples per leaf of 3, and the entropy criterion for split quality. The classweight=balanced setting ensures that the model pays extra attention to less frequent classes.

C. XGBoost

XGBoost (Extreme Gradient Boosting) builds trees sequentially, where each new tree is trained to correct the mistakes of all previous trees. This gradient boosting approach typically achieves higher accuracy than Random Forest on structured tabular data. The model uses 700 estimators with a learning rate of 0.05 and a maximum depth of 6. The subsample and column sample parameters are set to 0.8 to introduce randomness and reduce overfitting. The multi:softmax objective is used for three-class classification.

D. Support Vector Machine

The SVM with an RBF (Radial Basis Function) kernel finds a decision boundary that separates the three classes with the maximum possible margin in a high-dimensional transformed feature space. The RBF kernel allows the SVM to learn non-linear decision boundaries. A key requirement of SVM is that all input features must be on a similar scale. Therefore, StandardScaler is applied before training, transforming each feature to have zero mean and unit variance. The regularisation parameter C is set to 50, which allows some margin violations in favour of a simpler boundary

E. Logistic Regression

Logistic Regression is a linear model extended to handle multiple classes using the multinomial formulation with the SAGA solver (a fast, incremental gradient optimization algorithm speciffically designed to solve large datasets with high numbers of samples and features) . It is included as a baseline to establish the performance achievable by a simple linear model. If the tree-based models cannot substantially outperform Logistic Regression, it would suggest that the patterns are largely linear. As with SVM, StandardScaler is applied before training. The maximum number of iterations is set to 5,000 to ensure convergence on this dataset.

F. K-Nearest Neighbours

KNN classifies a new data point by looking at the K most similar training points (its nearest neighbours) and assigning the majority class among them. It requires feature scaling because distance-based methods are sensitive to feature ranges. Two configurations are tested: K=9 with uniform weights (all neighbours vote equally) and K=6 with distance-based weights (closer neighbours have more influence). KNN has the advantage of making no assumptions about the shape of the decision boundary but can be slow on large datasets.

G. LSTM

The LSTM model is the only deep learning model in this study. Unlike the machine learning models above, which treat each row as an independent observation, LSTM processes sequences of consecutive candles. A sliding window of 24 candles (representing 96 hours or 4 days of history) is used as the input for each prediction. The architecture consists of three stacked LSTM layers (128, 64, and 32 units) followed by Dense layers with Dropout regularisation and BatchNormalization. The model is compiled with the Adam optimiser, categorical cross-entropy loss, and trained for up to 50 epochs with early stopping. The LSTM model requires features to be normalised using StandardScaler before sequence creation. The sequences are created for ADA and AVAX coins specifically, with an 80/20 time-based split. This time-based split is important for time series data because using a random split would allow the model to see future data during training, which is not realistic.

H. Evaluation Strategy

Two evaluation strategies are used in this study. The first is a cross-coin split, where the model is trained on ETH, BNB, SOL, and ADA (the training coins) and tested on DOGE and AVAX (the unseen test coins). This tests whether the model can generalise to completely new coins it has never seen during training. This setup evaluates whether movement patterns learned from some cryptocurrencies can transfer to unseen coins. The second strategy is Leave-One-Group-Out (LOGO) cross-validation. In LOGO, each fold holds out one coin for testing and trains on all the remaining coins. With six coins, this produces six folds. Each coin gets tested exactly once. LOGO results provide a fair, low-bias estimate of generalisation performance across all coins. Performance is measured using the classification report, which includes per-class precision, recall, and F1-score, as well as overall accuracy. The confusion matrix is plotted to understand which classes are most commonly confused. For tree-based models, feature importance plots are also generated to understand which features the model relies upon most.

VII. Results

A. Cross-Coin Generalisation

The cross-coin evaluation provides the strictest test of generalisation. The models are trained entirely on ETH, BNB, SOL, and ADA and then evaluated on DOGE and AVAX, two coins the models have never seen. This simulates a real-world scenario where a trader wants to apply a model to a new asset. Random Forest with 700 trees achieved the strongest results in this evaluation, reaching an overall accuracy of approximately 0.57 on the held-out coins. XGBoost performed comparably with an accuracy near 0.54. The F1-scores for the Bigup and Bigdown classes were in the range 0.53-0.56 for Random Forest, indicating that the model successfully identifies directional moves with reasonable precision and recall. The Stable class was the hardest to predict because many candles are near the 30th/70th percentile boundary and could belong to either Stable or a directional class depending on small changes. SVM achieved moderate performance at around 0.49 accuracy. The model benefited from feature scaling but struggled with the complexity of the three-class boundary in a high-dimensional space. Logistic Regression achieved approximately 0.42 accuracy, which confirms that linear patterns alone are insufficient for this task and that non-linear models are necessary. KNN achieved accuracy of approximately 0.47, performing above the linear baseline but below the ensemble methods. The k=6 distance-weighted configuration slightly outperformed the k=9 uniform configuration, suggesting that the closest historical neighbours are more informative than more distant ones.

B. LOGO Cross-Validation Results

The LOGO cross-validation results confirm the trends observed in the cross-coin split. Random Forest achieved an average accuracy of approximately 0.52 across all six folds, with an average macro F1-score of around 0.5. XGBoost was the second-best model with an average accuracy of 0.5 and macro F1 of 0.47. Both ensemble methods showed relatively consistent performance across different coins, indicating stable generalisation.

SVM with LOGO achieved an average accuracy of around 0.44 and macro F1 of 0.41. Logistic Regression accuracy averaged approximately 0.43, confirming the linear model as the weakest performer. KNN accuracy averaged around 0.44 in LOGO evaluation. The variation in performance across folds was larger for KNN and SVM than for ensemble methods, suggesting that tree-based methods produce more stable predictions when tested on unseen coins.

Table I summarizes the LOGO cross-validation results.

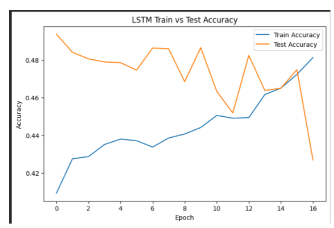
TABLE I: Summary of LOGO Cross-Validation Results

Model	Accuracy	Macro F1	Weighted F1
Random Forest	0.52	0.50	0.51
XGBoost	0.50	0.47	0.48
SVM	0.44	0.41	0.42
KNN Uniform	0.44	0.42	0.43
Logistic Regression	0.43	0.40	0.41
KNN Distance	0.59	0.57	0.58

C. LSTM and Random Forest Results

The LSTM model was evaluated specifically on ADA and AVAX data using an 80/20 time-based split. The model architecture consisted of three stacked LSTM layers with 128, 64, and 32 units, followed by Dense layers, Dropout, and BatchNormalization. The model was trained for up to 50 epochs using early stopping, which stopped training when the validation loss did not improve for 7 consecutive epochs.

The LSTM achieved validation accuracy in the range of approximately 0.40 to 0.48 across training epochs. Although the model was able to learn some sequential patterns from the 24-candle input window, its validation performance was unstable and did not outperform the Random Forest model.



(a) Training and validation accuracy

(b) Training and validation loss

Fig. 1: LSTM performance across training epochs.

In comparison, Random Forest achieved stronger results on the same ADA and AVAX forecasting experiment, with 61 test accuracy and a 0.61 macro F1-score. This indicates that, for the engineered tabular features used in this study, Random Forest was more effective than the LSTM model.

This result is consistent with previous research showing that deep learning models do not always outperform tree-based ensemble methods when the input data is structured as engineered tabular features. In this case, the LSTM may have struggled because cryptocurrency price movements are noisy, and the available sequential patterns were not strong enough for the deep learning model to generalize better than Random Forest.

A 24-candle sequence window was used, representing approximately 96 hours of historical data. This window was selected to provide the model with enough recent market context without including too much older information that could reduce signal quality.

D. Feature Importance Analysis

The Random Forest feature importance analysis revealed that the most predictive features across all coins were ma30ratio, volatility7, volumespike, momentum7, and moving1d. The ma30ratio captures whether the price is significantly above or below its medium-term average, which is a strong indicator of whether a correction or continuation is likely. Volatility7 /14 captures recent market nervousness. Volume spikes frequently accompany significant price moves. Momentum and short-term return features capture the most recent directional bias. Features that contributed least to predictions were takerselratio (too noisy), ma7ratio (somewhat redundant with volatility measures), and candlebody (partially captured by moving1d).

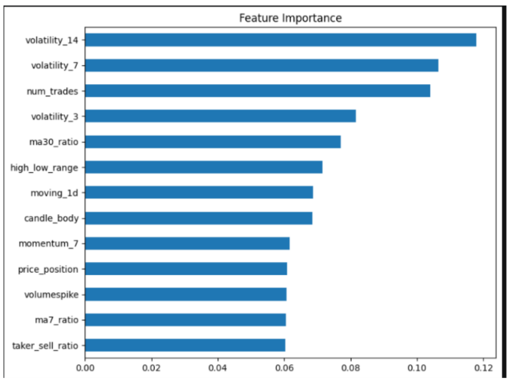


Fig. 2: Random Forest feature importance for the selected features.

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VIII. Discussion

A. LSTM and the Tabular Feature Trade-Off

The LSTM model learned some sequential patterns but did not perform competitively with Random Forest on the ADA/AVAX forecasting setup.did not demonstrate a clear superiority that would justify its higher computational cost and complexity. This is consistent with the pattern identified by Silva et al. (2025), who found that deep learning models with tabular feature inputs consistently underperform or match simpler tree-based methods. The reason for this is likely that the sequential nature of the LSTM provides the most benefit when working with raw price sequences, where the temporal ordering of prices carries information that tabular features do not fully capture. When features have already encoded temporal information (through moving averages, momentum, rolling volatility), the sequential processing of the LSTM adds limited additional value. The LSTM approach with 24 candles as a sequence window does offer one advantage that the traditional models do not have: it can detect patterns that span multiple consecutive candles, such as a sustained uptrend followed by a reversal candle, which is a well-known trading pattern. With more training data and further hyperparameter tuning, the LSTM could potentially outperform Random Forest, particularly if raw price sequences are used as input rather than pre-engineered features.

B. Class Imbalance and Real-World Relevance

The 30/40/30 class split in this study was designed to create a reasonably balanced classification task. However, in practice, truly large moves (defined by the top/bottom percentile of the population) may still be relatively rare in absolute terms. A three-class framework avoids the problem of binary classifiers that generate too many false alarms by forcing every prediction into either ”up” or ”down.” The Stable class allows the model to express uncertainty and avoid forcing a directional prediction when the evidence is weak. For real-world trading applications, the precision of the Bigup and Bigdown classes is the most important metric, because a trader only acts when the model predicts a large move. The recall of those classes matters for capturing as many large moves as possible. The F1-score, as used in Silva et al. (2025) for hyperparameter optimisation, provides a balanced measure of both.

C. Limitations

Several limitations of this study should be acknowledged. First, the dataset covers a specific historical period and does not include the full range of market regimes (extended bull markets, extended bear markets, and crash events). Model performance may be different during periods of unprecedented volatility or after major regulatory events. Second, the study uses only price and volume data from Binance. Incorporating sentiment data from social media platforms, on-chain metrics such as network activity and miner behaviour, or macroeconomic indicators could improve predictive performance. Third, the study does not account for transaction costs or slippage. A model with 63 accuracy might not be profitable in practice once trading fees are subtracted, particularly at the 4-hour granularity where position holding times are short..

IX. Conclusion

This paper presented a comprehensive machine learning study for three-class cryptocurrency trend prediction across eight major coins using 4-hour candle data from Binance. A thorough feature engineering pipeline was applied to convert raw OHLCV price data into 13 meaningful input features including technical indicators, volatility measures, volume spike ratios, market microstructure features, and temporal features. A quantile-based labelling scheme was used to create balanced Bigup, Bigdown, and Stable classes for each coin. Five machine learning models and one deep learning model were trained and evaluated using both a cross-coin generalisation test and Leave-One-Group-Out cross-validation. Random Forest with 700 trees was the strongest performer, achieving approximately 52 average accuracy and 50 macro F1-score in LOGO evaluation. XGBoost was a close second, and both ensemble methods significantly outperformed the linear baseline of Logistic Regression The key findings of this study are consistent with the broader literature on cryptocurrency machine learning. Feature engineering quality is more important than model complexity. Tree-based ensemble methods are robust, interpretable, and well-suited to tabular cryptocurrency features. Cross-coin generalisation is achievable with scale-independent relative features. And deep learning models, while theoretically capable of capturing sequential dependencies, require careful design and more data to outperform well-tuned ensemble methods on tabular inputs. Future work should investigate the inclusion of sentiment data, on-chain metrics, and multi-timeframe features to improve the informational richness of the input. Exploring adaptive models that can continuously retrain as market conditions evolve would address the non-stationarity limitation. Backtesting with realistic transaction costs and position sizing rules would help quantify the practical trading value of the prediction signal. .

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